



Cloud Computing

- The internet is often represented as a “**cloud**”, because users don’t need to know the exact location of data or services.
- **Cloud computing** means:
Providing data storage, applications, and services to many users over a network, in a flexible and scalable way.

- **Key features of cloud computing:**



- **On-demand service:** Users can get resources whenever they need.
- **Elasticity:** Storage and services can grow or shrink based on usage.
- **Multi-platform support:** Works with many operating systems and devices.
- **Resource pooling:** Many users share the same infrastructure efficiently.
- **Metered service:** You pay only for what you use (like electricity or water).

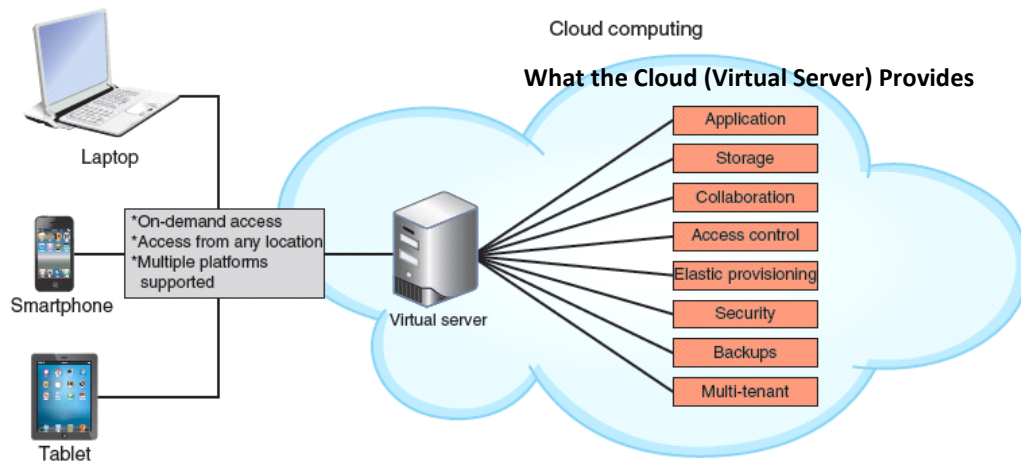


Figure 7-14 Characteristics of cloud computing

- Cloud services can offer **virtual desktops**, meaning users work in operating systems hosted online instead of on local machines.
- **Developers** can remotely install and test any software on cloud servers. The cloud provider ensures that these servers are **secure** and **backed up regularly**.
- Most providers use **virtualization software** to deliver different platforms (like Windows, Linux) to many users at the same time, using a single physical server.



Cloud Computing Service Models

Cloud computing is divided into different **service models** based on what kind of services are offered:

1. IaaS (Infrastructure as a Service)

→ Provides **hardware** like servers, storage, and networking over the internet.

Example: You rent virtual machines but manage your own software.

2. PaaS (Platform as a Service)

→ Offers an environment with **OS and runtime tools** so developers can build and run applications without managing the infrastructure.

Example: The provider gives you everything you need to run your code.

3. SaaS (Software as a Service)

→ Delivers **ready-to-use software applications** over the web.

Example: Gmail, Microsoft 365, or Dropbox.

4. XaaS (Anything as a Service)

→ A broad term that means **any IT function** (like databases, analytics, etc.) can be offered as a cloud service.



Cloud Deployment Models

Cloud services can be deployed in different ways :

1. **Public Cloud**
 - Services are offered over the **public internet** by companies like Google or Amazon.
 - Anyone can use them by paying a fee.
2. **Private Cloud**
 - The organization **builds and manages its own cloud** inside its own data center.
 - More **control and security**.
3. **Community Cloud**
 - Used by **several organizations** with similar needs (e.g., universities or government agencies).
 - They share the same cloud system.
4. **Hybrid Cloud**
 - A **mix** of two or more types (usually public + private).
 - Allows companies to keep sensitive data in a private cloud and use public cloud for general tasks.



Remote Access

- **Remote access** allows a user to connect to a server, LAN, or WAN from another location. This is useful for people working from home or traveling.



Types of Remote Access:

1. **Point-to-point**: Uses a special connection line between two locations.
2. **VPN (Virtual Private Network)**: Creates a secure tunnel through the internet.
3. **Remote terminal emulation**: Lets you control another computer as if you were sitting at it.



Remote Access Server (RAS)

- A **Remote Access Server (RAS)** accepts incoming remote connections and gives access to network resources.



Two Types of RAS:

1. **Dedicated hardware devices**
 - Example: Cisco AS5800
2. **Software-based servers**
 - Example: Microsoft DirectAccess

How It Works

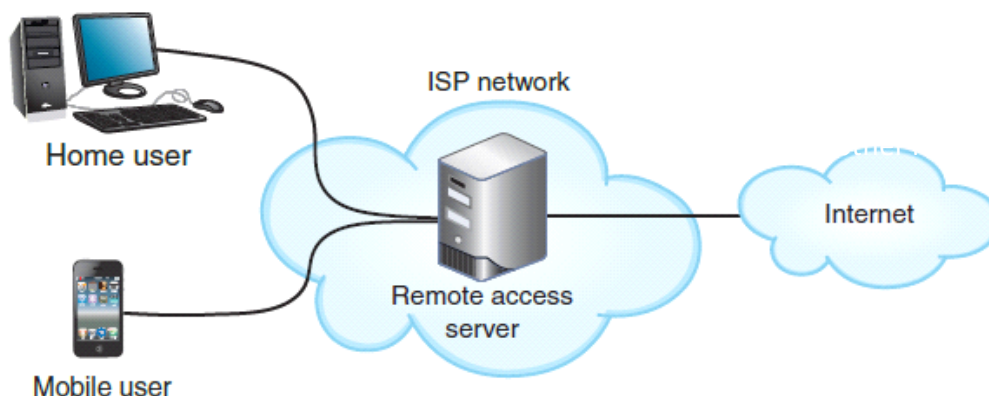


Figure 7-19 An ISP uses a remote access server to authenticate subscribers to its services, including access to the Internet

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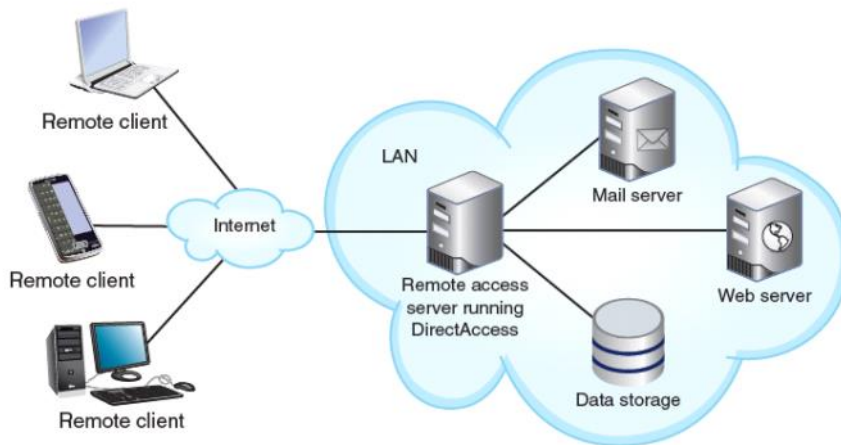


Figure 7-20 DirectAccess authenticates users to the Windows domain

- **Remote clients** (laptop, mobile, PC) use the **internet** to reach the company's **Remote Access Server**.
- **DirectAccess authenticates** (verifies) the users.
- Once authenticated, users can securely access:
 - **Mail server**
 - **Web server**
 - **Data storage**

remote users (clients) can connect to a company's private network using the **internet**.

! Point-to-Point Remote Access Protocols

1. SLIP (Serial Line Internet Protocol)

- **Older** and more **basic** protocol than PPP.
- It can **only carry IP packets**, and it works **only over serial connections** (cable-based).
- It has no encryption or advanced features.

! 2. PPP (Point-to-Point Protocol)

- More **advanced** and flexible than SLIP.
- Can **create a connection** between two computers. !!!
- Can **authenticate** the user (check their identity).
- Supports **different network layer protocols**, not just IP.
- Can **encrypt** data, but its encryption is now considered **weak** by modern standards.

VPN (Virtual Private Networks)

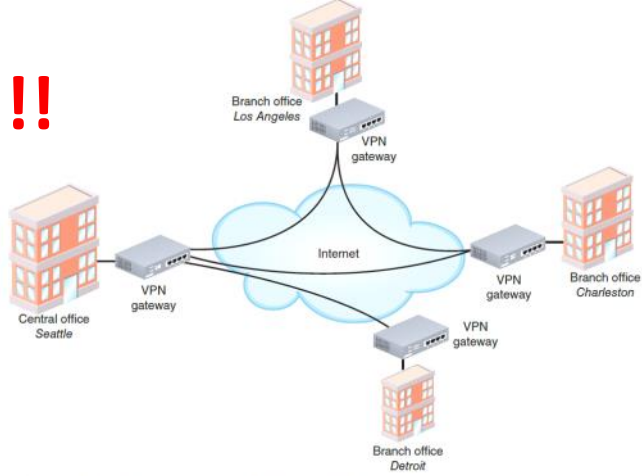
- A **VPN** is a **virtual network** that allows **secure communication** over a public internet connection.
- It creates a **private tunnel** inside a public network, so data stays safe.

How it works:

- VPNs use **special protocols** to wrap (or encapsulate) data in a process called **tunneling**.
- This lets VPNs carry all types of data securely, even over unsafe networks.

! VPN Types:

1. **Site-to-site VPN** → Connects two offices or networks together.
2. **Client-to-site VPN** → Also called **host-to-site** or **remote-access VPN**.



- The diagram shows how a **Site-to-Site VPN** works.
- There is one **central office** (Seattle) and three **branch offices** (Los Angeles, Detroit, Charleston).
- All offices are connected over the **public internet** using **VPN gateways**.

Figure 7-24 A VPN gateway connects each site to one or more other sites

1. Each user connects through the **internet** to the **VPN gateway** at the Charleston office.
2. The gateway gives them **secure access** to the company's LAN (internal network).
3. This way, they can work remotely as if they were in the office.

Example of Client-to-Site VPN

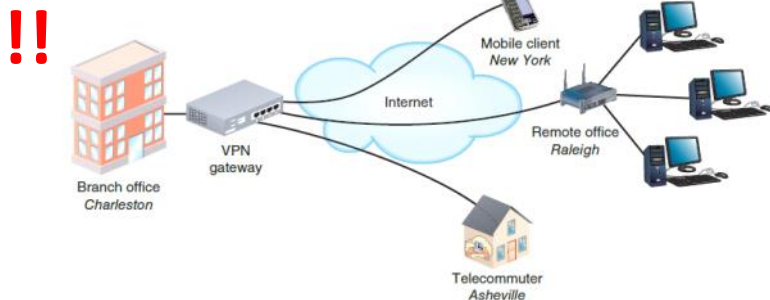


Figure 7-25 Remote clients connect to the LAN through the VPN gateway

How VPNs Are Installed or Managed

There are different ways to **set up and run a VPN**:

1. **Built-in VPN software (embedded the OS)**
 - Example: **RRAS** (Routing and Remote Access Service), which is Microsoft's built-in VPN and remote access tool.
2. **Third-party software**
 - Example: **OpenVPN** is **open-source** and works on many platforms.
3. **Routers or firewalls**
 - Some VPNs are built into **network hardware** like routers or firewalls.
 - Most common implementation of VPNs on **UNIX-based networks**.

VPN Concentrator & Encryption

- A **VPN concentrator** is a **special network device**:
 - It **authenticates** VPN users (checks who they are),
 - **Creates secure tunnels** for communication,
 - **Encrypts data** going through the VPN.
 - It is also called an **encryption device**.

VPN Encryption Types

VPNs commonly use **two main encryption methods**:

1. **IPsec** – Used for secure IP communications.
2. **SSL** – Often used in browser-based VPNs (like HTTPS websites).

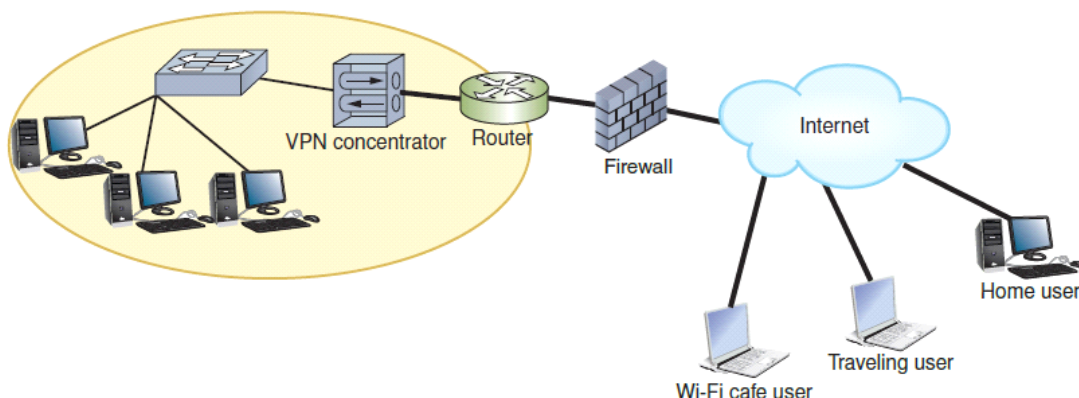


Figure 7-26 Placement of a VPN concentrator on a LAN

How It Works (Step by Step):

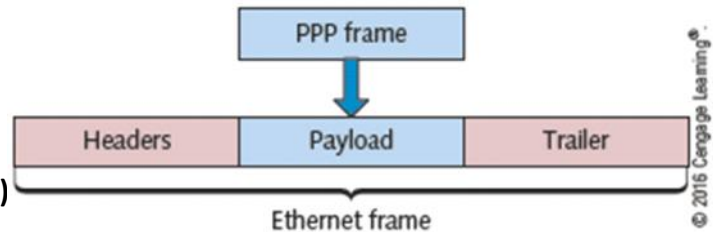
1. **Home user, Wi-Fi café user, and traveling user** all connect to the **internet**.
2. From there, they reach the **company's firewall** and pass through it if allowed.
3. Their connection goes to the **VPN concentrator**, which:
 - Authenticates them,
 - Establishes a secure **VPN tunnel**,
 - Encrypts and decrypts data.

4. After that, they can safely access the **LAN (Local Area Network)**, just like employees inside the company building.



VPN Tunneling Protocols

- VPN **tunneling protocols** are used to send VPN data **securely** over the internet.
- These protocols work at the **Data Link Layer** of the OSI model.
- The VPN data (called a **PPP frame**) is **encapsulated into Network packet**.
- This packet becomes part of an **Ethernet frame**, which includes:
 - **Headers** (info about where it's going),
 - **Payload** (the actual data),
 - **Trailer** (error-checking info).



Common Tunneling Protocols:

1. PPTP (Point-to-Point Tunneling Protocol)
2. L2TP (Layer 2 Tunneling Protocol)

Figure 7-8 The VPN frame is encapsulated inside the Network layer packet

PPTP (Point-to-Point Tunneling Protocol)

- A Layer 2 protocol that **wraps PPP data** so it can be sent over the internet as if it were normal IP traffic.
- It uses **TCP** at the transport layer.
- Often used with **GRE (Generic Routing Encapsulation)**, it helps carry PPP frames inside IP packets.

PPTP is **no longer considered secure**, L2TP recommended today.

GRE (Generic Routing Encapsulation)

- GRE is used to **send PPP data** through a VPN tunnel.
- It **wraps PPP data** so it can look like a regular IP packet while moving across the internet.

L2TP (Layer 2 Tunneling Protocol)

- Works **similarly to PPTP**, but is **more secure and reliable**.
- Can connect different types of devices (routers, clients, RAS).
- It's a **standard** supported by many network providers.
- Useful in environments with **mixed equipment**.

Terminal Emulation (Remote Virtual Computing)

- Allows a someone use one computer to work on another computer through a network.

◆ Command-line tools:

– Telnet, SSH

◆ GUI tools:

– Remote Desktop, join.me, VNC, TeamViewer

Encryption Techniques, Protocols, and Utilities



Encryption

- Encryption uses **math-based codes (ciphers)** to change data into unreadable form.
- Only someone with the correct method (key) can **decode** it.
- It keeps data **private** and ensures:
 - **Confidentiality** (only authorized people can read it)
 - **Integrity** (data is not changed)
 - **Availability** (data is accessible when needed)

Key Encryption

- A **key** is a random string used to **change original data** into a unique form.
- Encrypted data is called **ciphertext** (it looks scrambled).
- A **brute-force attack** is when someone tries many key combinations to guess the correct one.

Private Key Encryption

- Uses **one key** for both **encrypting** and **decrypting** data.
- The key is **shared only between** the sender and the receiver.
- Also called **symmetric encryption**.

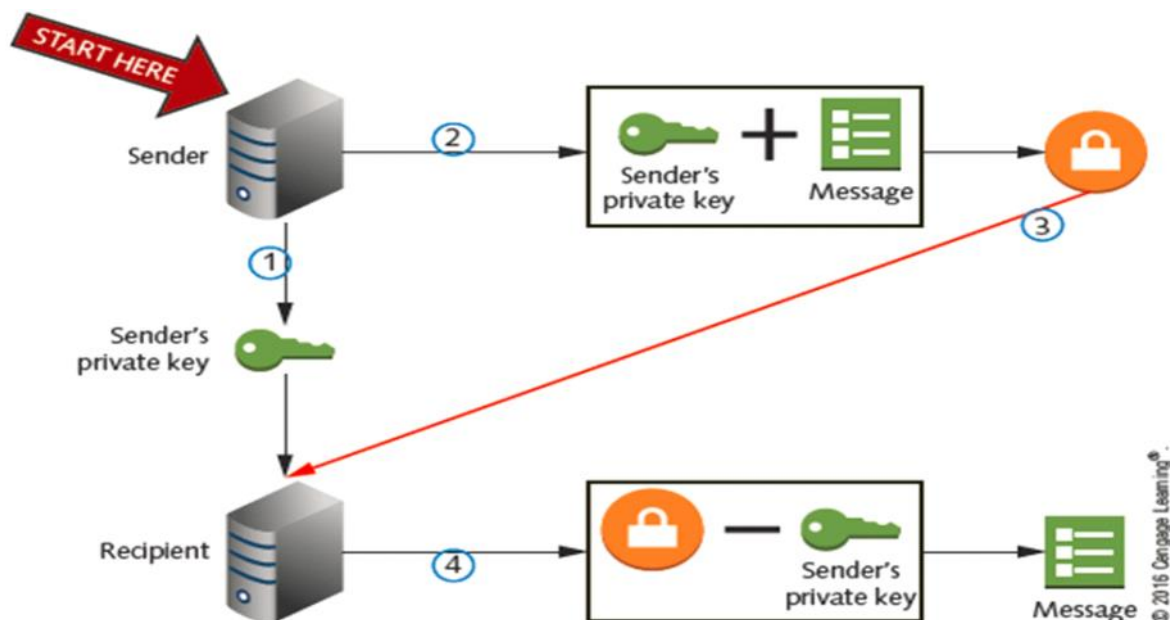


Figure 7-10 Private key encryption begins with the sender

- The **sender** uses a **private key** to encrypt the message.
- The message is now locked and unreadable.
- The locked message is sent to the **recipient**.
- The **recipient** uses the **same private key** to unlock (decrypt) the message.

KEY ENCRYPTION !**Public key encryption**

- Uses **two keys** to protect data
- **Private key** is kept secret by the user
- **Public key** can be shared with anyone

Public key server

- A **public platform** that stores public keys
- Anyone can **freely access** these keys

Key pair

- A combination of **public** and **private** key
- These two keys work **together** for encryption and decryption

Asymmetric encryption

- Requires **two different keys**, same key is **not used** for both operations

“Draw and explain the steps of public key encryption.

Show how the sender uses the recipient’s public key to send a secure message, and how the recipient uses their private key to decrypt it.”

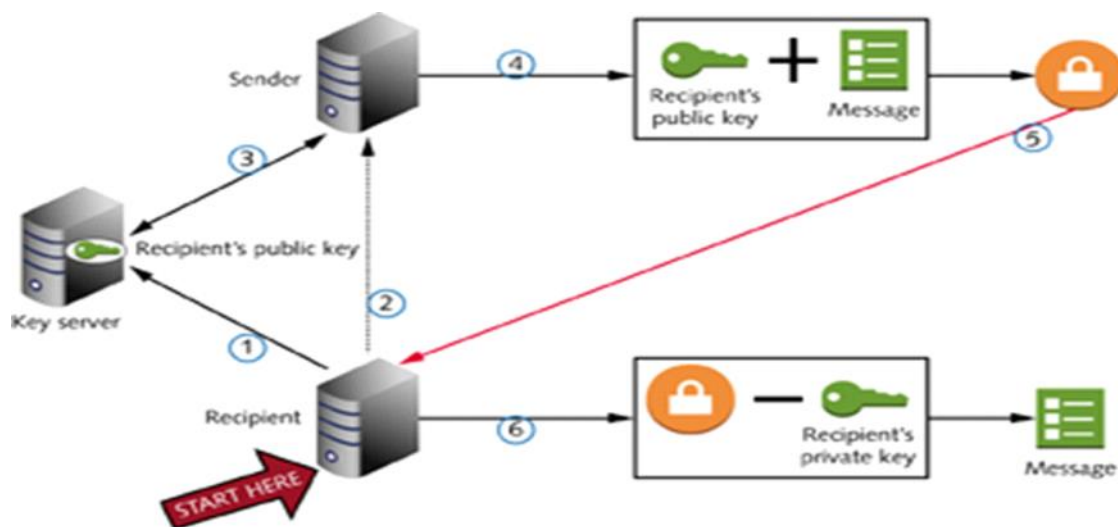


Figure 7-11 Public key encryption begins with the recipient

1. The **recipient** gets a **public key** from the **key server**.
 2. The recipient keeps their **private key** secret.
 3. The **sender** also gets the recipient’s **public key** from the server.
 4. The sender uses that public key to **encrypt the message**.
 5. The **encrypted message** is sent to the recipient.
 6. The recipient uses their **private key** to **decrypt** the message.
- ✓ Only the private key can unlock the message.

Key Encryption !

- **Digital certificate**
 - Stores user identity information
 - Contains the user's public key
- **CA (Certificate Authority)**
 - checks identity and signs the certificate, ensuring it’s trustable
- – Example: Verisign
- **PKI (Public Key Infrastructure)**
 - A system that uses CAs to connect **public keys** with specific **users**

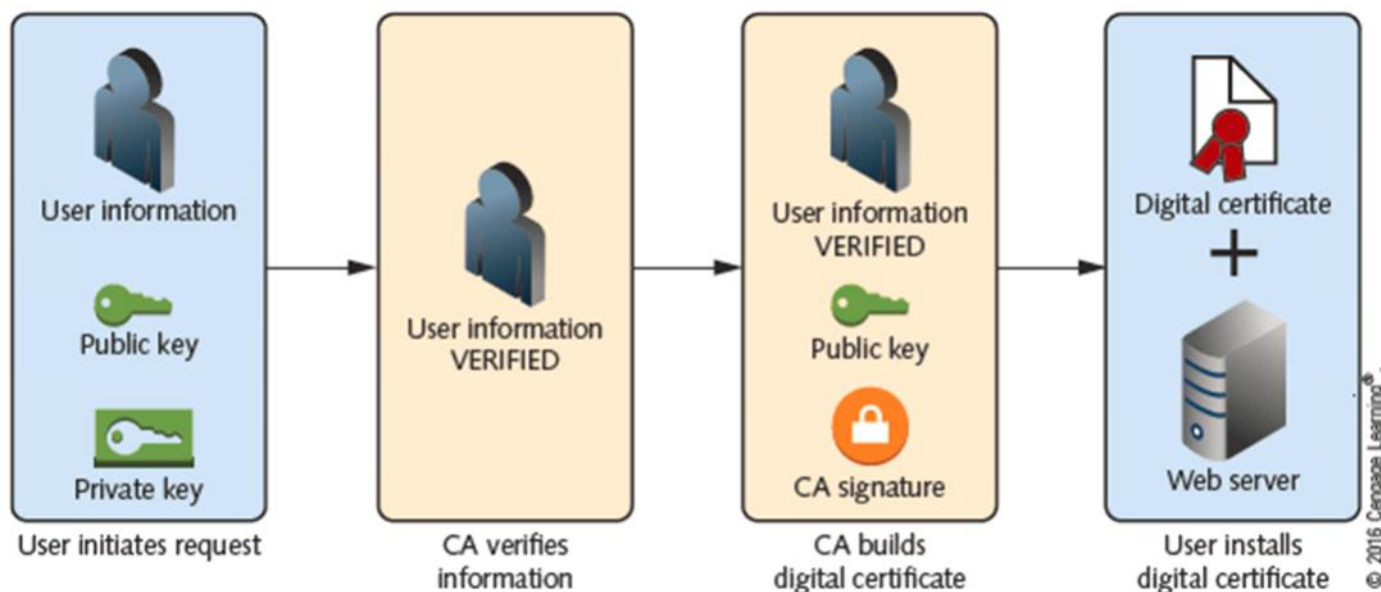
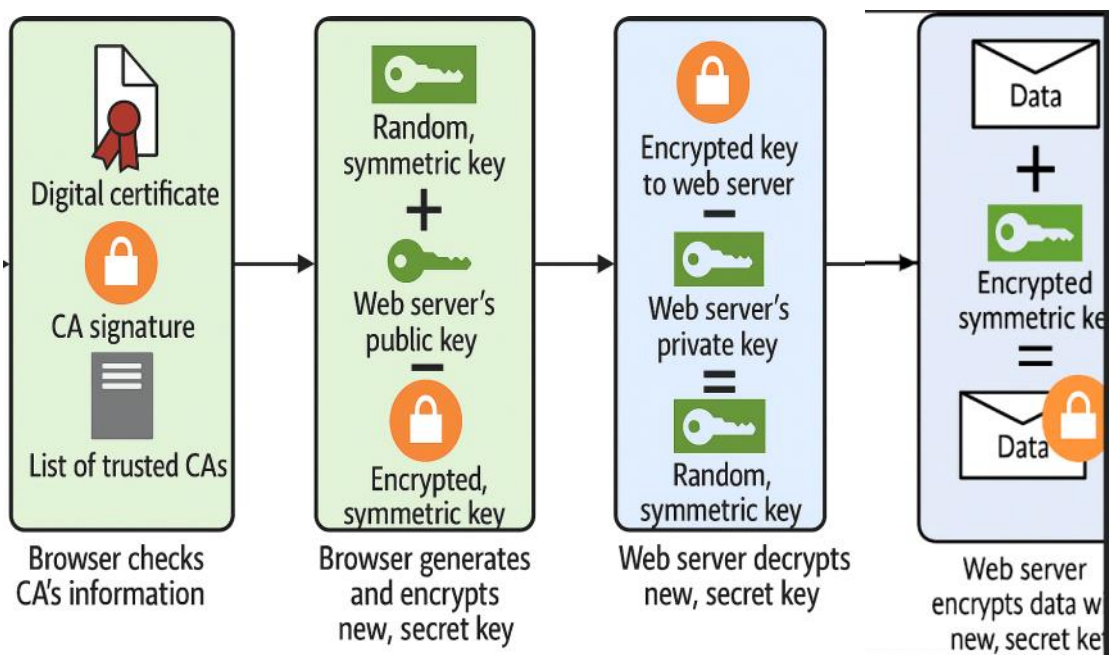


Figure 7-12 It's important to obtain digital certificates only from trusted CAs

Digital Certificate Process

1. The **user** sends their information, public key, and private key to request a certificate.
2. The **Certificate Authority (CA)** checks the user's information.
3. If it's correct, CA **adds its signature** and builds a **digital certificate** with the public key.
4. The user then **installs this certificate** on their **web server**.

✓ This process proves the user is trusted and owns the public key.



How a Secure Connection Works (SSL/TLS - Simplified)

- **First**, the browser checks the server's **digital certificate** signed by a trusted **CA**.
- Then it creates a **random symmetric key**.
- This symmetric key is **encrypted using the server's public key** → (public key encryption used).
- The server decrypts it with its private key.
- Now both sides have the same symmetric key, and **all future data is encrypted using that key** → (private key encryption starts).

Private Key Encryption (Symmetric Encryption)

- Uses **one single key** for both **encrypting and decrypting** data.
- The same key is **shared between the sender and the receiver**.
- Known only by both sides.
- It is **faster** than public key encryption.
- It is used **after the connection is made**, to protect the **main data transfer**.



Public Key Encryption

- Uses **two keys**: a **public key** (anyone can use it) and a **private key** (only the owner knows).
- Data is **encrypted with the receiver's public key**, and only the **receiver's private key** can decrypt it.
- It is used **at the beginning of communication** to **securely send the symmetric key**.